

Lab Final Project Instructions

Submission Reminder

Do not forget to submit your project by **11:59 PM on the due date**, via Gradescope.

Project Overview

This Lab Final Project is graded out of 10 points. You are required to perform the following steps:

1. **Pick a dataset.** Select an appropriately large dataset from one of the repositories listed below, or any other source of interest. Requirements:
 - **High-dimensional:** The number of attributes/features should be large and proportional to the number of samples/instances.
 - **Aspect ratio:** Look for $\gamma = \frac{\text{\#features}}{\text{\#samples}}$ between 1/20 and 20.
 - If unsure, consult the instructor or TA.
2. **Compute eigenvalues.** Calculate the eigenvalues of the sample covariance matrix and plot a histogram. Fit the **best Marčenko-Pastur density** (with the appropriate γ) to determine the **empirical noise edge**.
3. **Identify outliers.** Determine which eigenvalues are outside the predicted Marčenko-Pastur bulk.
4. **Principal components.** Using PCA/SVD, find the principal components corresponding to the outlier eigenvalues.
5. **Project data.** Plot the projection of the data onto these principal coordinates.
 - If $d > 3$, consider subsets of 2–3 principal components. Comment on each projection.
6. **Analyze and interpret.** Examine the dimension-reduced data statistically or descriptively. Notes:
 - **Strong correlations:** The Marčenko-Pastur fit may be poor.
 - **Sparse data:** Many noise-induced outliers may appear.
 - **Important:** The largest outliers are usually statistically significant.

Format of Submission

You may submit in **either** of the following formats:

- A **typed report in L^AT_EX**, with:
 - Step-by-step figures
 - Explanations of methodology
 - Analysis of results
 - Code included or in an appendix
- A **notebook** (like Labs) with descriptions, commentary, analysis, and methodology included.

References: Provide a citation for your dataset at the start of the report.

Submission format: PDF only, uploaded to Gradescope.

Recommended Aspect Ratios

Here is a quick reference for acceptable dataset dimensions:

#Samples (n)	#Features (m)	Aspect Ratio $\gamma = m/n$
1000	50	1/20
500	10	1/50 (too low, consult TA)
200	200	1
100	2000	20

Suggested Dataset Repositories

You may use any dataset satisfying the above criteria. Suggested repositories include:

- **Data Nexus (UCSD)** — Links to 3 repositories of Machine Learning datasets (scroll down the page).
- **UCI Machine Learning Repository**.
- **OpenML** — Dataset dimensions are displayed for easy selection.